

AST 6007-01 Spacecraft Dynamics and Control

Final Examination

December 2, 2015 1:00-3:00 PM

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In the beginning of your answer sheet, sign the honor code: I never received nor gave help regarding this exam.

1. (20) Suppose that a spacecraft with mass $m=100\text{kg}$ operates near a circular orbit with orbital radius $r=6,578\text{km}$. The Clohessy-Wiltshire equations can describe its orbital perturbation dynamics in the LVLH $\hat{i}\hat{j}\hat{k}$ -frame. Under some conditions, the Clohessy-Wiltshire equations can be further approximated by

$$\begin{cases} m\Delta\ddot{x} = F_x \\ m\Delta\ddot{y} = F_y \\ m\Delta\ddot{z} = F_z \end{cases}$$

Here $(\Delta x, \Delta y, \Delta z)$ are perturbations of the spacecraft position from the circular orbit, and (F_x, F_y, F_z) are propulsive force components expressed in the LVLH $\hat{i}\hat{j}\hat{k}$ -frame. This approximation is often acceptable over a sufficiently small time period $T > 0$.

- a. (10) Estimate how small the maneuver time period T should be so that the above equations are a satisfactory approximation of the Clohessy-Wiltshire equations within accuracy of about 3%. Hint: Consider the following analytic solutions to the Clohessy-Wiltshire equations:

$$\begin{aligned} x(t) &= \frac{\dot{x}_0}{\omega} \sin(\omega t) - \left(3x_0 + \frac{2\dot{y}_0}{\omega} \right) \cos(\omega t) + \left(4x_0 + \frac{2\dot{y}_0}{\omega} \right) \\ y(t) &= \left(6x_0 + \frac{4\dot{y}_0}{\omega} \right) \sin(\omega t) + \frac{2\dot{x}_0}{\omega} \cos(\omega t) - (6\omega x_0 + 3\dot{y}_0)t + \left(y_0 - \frac{2\dot{x}_0}{\omega} \right) \\ z(t) &= \frac{\dot{z}_0}{\omega} \sin(\omega t) + z_0 \cos(\omega t) \end{aligned}$$

$x(t) - x_0 < 0.03x(t)$
 $y(t) - y_0 < 0.03y(t)$

- b. (10) Using the above approximation, a two-impulse rendezvous maneuver is to transfer the initial spacecraft position vector $\Delta\vec{r}(0) = -10\hat{i}\text{km}$ and initial velocity vector $\Delta\dot{\vec{r}}(0) = 0$ to the terminal spacecraft position vector $\Delta\vec{r}(60\text{sec}) = 0$ and terminal velocity vector $\Delta\dot{\vec{r}}(60\text{sec}) = 0$. What is the change in spacecraft velocity vector that occurs at each of the impulsive burns?

2. (20) Consider the equations of motion for the Earth-Moon circular restricted three-body problem:

$$\begin{cases} \ddot{x} - 2\omega_0 \dot{y} - \omega_0^2 x = -\frac{\mu_E(x - D_E)}{r_E^3} - \frac{\mu_M(x + D_M)}{r_M^3} + \frac{F_x}{m} \\ \ddot{y} + 2\omega_0 \dot{x} - \omega_0^2 y = -\frac{\mu_E y}{r_E^3} - \frac{\mu_M y}{r_M^3} + \frac{F_y}{m} \\ \ddot{z} = -\frac{\mu_E z}{r_E^3} - \frac{\mu_M z}{r_M^3} + \frac{F_z}{m} \end{cases}, \begin{cases} r_E^2 = (x - D_E)^2 + y^2 + z^2 \\ r_M^2 = (x + D_M)^2 + y^2 + z^2 \\ \mu_E = 398,601 \text{ km}^3 / \text{sec}^2 \\ \mu_M = 4,887 \text{ km}^3 / \text{sec}^2 \\ D_E = 4,674 \text{ km} \\ D_M = 380,073 \text{ km} \\ \omega_0 = 2.66199 \times 10^{-6} \text{ rad/sec} \end{cases}$$

- a. (10) Linearize the above equations of motion with respect to the Lagrangian point $L_2: (\bar{x} = -444,645.6 \text{ km}, \bar{y} = 0, \bar{z} = 0)$.
- b. (5) What are the eigenvalues of the linearized out-of-plane equations of motion?
- c. (5) Discuss the stability of out-of-plane motion based on its eigenvalues.

3. (30) Euler's Equations of Motion

- a. (15) Starting from Newton's law that the inertial time rate of change of the angular momentum is the total applied torque, derive Euler's equations of motion in a body fixed frame, which is assumed to be principal. The moment of inertia in the given set of principal axes, the angular velocity component and the external moment component are given as $I = \text{diag}(A, B, C)$, $\vec{M} = [M_1 \ M_2 \ M_3]^T$ and $\vec{\omega} = [\omega_1 \ \omega_2 \ \omega_3]^T$, respectively.
- b. (15) In case the rotational motion is torque-free, show that the kinetic energy and the magnitude of angular momentum are conserved.

4. (30) Consider the pitch axis control model for a spacecraft in a circular orbit given by

$$I_y \Delta \ddot{\theta} = 3\omega_0^2 (-I_x + I_z) \Delta \theta + M_y$$

where the orbital frequency $\omega_0 = 0.0012 \text{ rad/sec}$, the spacecraft principal moments of inertia are $I = \text{diag}(150, 100, 200) \text{ kg-m}^2$. In this equation, $\Delta \theta$ denotes the pitch angle and M_y denotes the pitch axis control moment.

- a. (15) If the pitch axis control moment $M_y = 0$, show that the pitch equilibrium $(\Delta \theta, \Delta \dot{\theta}) = (0, 0)$ is an unstable equilibrium.
- b. (15) Show that linear state feedback to the pitch axis control moment can be used to stabilize the pitch axis equilibrium $(\Delta \theta, \Delta \dot{\theta}) = (0, 0)$ of the spacecraft. Choose feedback gains so that the equilibrium of the closed loop is asymptotically stable with damping ratio of 0.7 and natural frequency of 0.008 rad/sec (that is, the closed loop characteristic polynomial is $s^2 + 0.0112s + 0.000064$).